

Education	PhD, Computer Science Saarland University Max Planck Institute for Software-Systems (MPI-SWS), Saarbruecken, Germany Thesis: Human-in-the-loop Specification-Driven Automation for Cloud System Design and Optimization Advisors: Antoine Kaufmann, Deepak Garg, (past: Jonathan Mace)	2020-current
	MSc, Computer Science University of British Columbia, Vancouver, Canada Thesis: Dara the explorer: coverage based exploration for model checking of distributed systems in Go. Advisor: Ivan Beschastnikh, (2nd Reader: Margo Seltzer)	2018-2020
	BSc, Computer Science University of British Columbia, Vancouver, Canada	2013-2018
Employment	Research Intern, Microsoft Azure Research - Systems - Designed and Developed Cerulean, a tool for automatically generating microservice systems from natural language descriptions. - Mentors: Rodrigo Fonseca, Alok Gautam Kumbhare - Collaborators: Pedro Las-Casas, Celine Irvine, Jonathan Mace, Gagan Somashekar, Chetan Bansal, Ricardo Bianchini	2024
	Research Intern, MPI-SWS Instrumented DeathStarBench applications with XTrace tracing for large-scale experiments for Sifter.	2019
	Undergraduate Research Assistant University of British Columbia, Under Ivan Beschastnikh Designed and developed Dara, a tool for model checking distributed systems.	2018
	Software Engineering Intern, NVIDIA MODS (Modular Diagnostics) Team - Implemented memory repair sequences for faulty High Bandwidth Memory(HBM). - Designed, developed, and implemented a CUDA based full memory stress test for identifying faulty HBM cells.	2017
	Software Engineering Intern, NVIDIA MODS (Modular Diagnostics) Team Implemented a synchronization option for CUDA based linpack tests to synchronize CUDA kernel launches within 30μs across multiple GPUs.	2016
	Software Developer Intern, Thinkbox Software Sequoia Team Designed, developed, and implemented the 3D PDF export feature for point cloud meshes in Sequoia.	2015-2016

Publications

Peer-Reviewed Publications

1. Iridescent: A Framework Enabling Online System Implementation Specialization. **Vaastav Anand**, Deepak Garg, Antoine Kaufmann. Under Review.
2. Epistemic Observability in Predictive Systems: An Impossibility Result. Tony Mason, **Vaastav Anand**. Under Review.
3. Peeking Under the Hood of Multi-Agent Systems. Tie Ma*, Yixi Chen*, **Vaastav Anand**, Alessandro Cornacchia, Amandio R. Faustino, Guanheng Liu, Shan Zhang, Hongbin Luo, Suhaib A. Fahmy, Zafar A. Qazi, Marco Canini. 1st ACM Conference on AI and Agentic Systems, System Demonstrations Track, **CAIS'26 Demos**.
4. Intent-Based System Design and Operation. **Vaastav Anand**, Yichen Li, Alok Gautam Kumbhare, Celine Irvine, Chetan Bansal, Gagan Somashekar, Jonathan Mace, Pedro Las-Casas, Ricardo Bianchini, Rodrigo Fonseca. Practical Adoption Challenges of ML for Systems (co-located with SOSP'25), **PACMI'25**.
5. DMAS-Forge: A Framework for Transparent Deployment of AI Applications as Distributed Systems. Alessandro Cornacchia, **Vaastav Anand**, Muhammad Bilal, Zafar Ayyub Qazi, Marco Canini. 1st Workshop on Systems for Agentic AI (co-located with SOSP'25), **SAA'25**.
6. Generating representative macrobenchmark microservice systems from distributed traces with Palette. **Vaastav Anand**, Matheus Stolet, Jonathan Mace, Antoine Kaufmann. 16th ACM SIGOPS Asia-Pacific Workshop on Systems (co-located with SOSP'25), **ApSys 2025**.
7. Automated Service Design with Cerulean (Project Showcase). **Vaastav Anand**, Alok Kumbhare, Celine Irvine, Chetan Bansal, Gagan Somashekar, Jonathan Mace, Pedro Las-Casas, Rodrigo Fonseca. 6th International Workshop on Cloud Intelligence/AIOps Workshop (co-located with ICSE), **AIOps 2025**.
8. Towards Using LLMs for Distributed Trace Comparison (Abstract). **Vaastav Anand**, Pedro Las-Casas, Rodrigo Fonseca, Antoine Kaufmann. 6th International Workshop on Cloud Intelligence/AIOps Workshop (co-located with ICSE), **AIOps 2025**.
9. Blueprint: A Toolchain for Highly Reconfigurable Microservice Applications. **Vaastav Anand**, Deepak Garg, Antoine Kaufmann, Jonathan Mace. In Symposium on Operating Systems Principles, **SOSP 2023**.
10. The Benefit of Hindsight: Tracing Edge Cases in Distributed Systems. Lei Zhang, Zhiqiang Xie, **Vaastav Anand**, Ymir Vigfusson, Jonathan Mace. In Networked Systems Design and Implementation, **NSDI 2023**.
11. The Odd One Out: Energy is not like Other Metrics. **Vaastav Anand**, Zhiqiang Xie, Matheus Stolet, Roberta De Viti, Thomas Davidson, Reyhaneh Karimipour, Safya Alzayat, Jonathan Mace. In **HotCarbon 2022**.
12. Smooth Kronecker: Solving the Combing Problem in Kronecker Graphs. **Vaastav Anand***, Puneet Mehrotra*, Daniel Margo*, Margo Seltzer. In Joint Workshop on Graph Data Management Experiences and Systems and Network Data Analytics, **GRADES-NDA 2020**.
13. Sifter: Scalable Sampling for Distributed Traces, without Feature Engineering. Pedro Las-Casas, Giorgi Papakerashvili, **Vaastav Anand**, Jonathan Mace. In Symposium on Cloud Computing, **SoCC 2019**.

Pre-Prints

1. MAESTRO: Multi-Agent Evaluation Suite for Testing, Reliability, and Observability. Tie Ma*, Yixi Chen*, **Vaastav Anand**, Alessandro Cornacchia, Amanda R. Faustino, Guanheng Liu, Shan Zhang, Hongbin Luo, Suhaib A. Fahmy, Zafar A. Qazi, Marco Canini. In **arXiv 2026** <https://arxiv.org/pdf/2601.00481>.
2. SoK: Demystifying the multiverse of MPC protocols. Roberta De Viti, **Vaastav Anand**, Pierfrancesco Ingo, Deepak Garg. In **arXiv 2025** <https://arxiv.org/pdf/2512.11699>.
3. Chatting with Logs: An exploratory study on Finetuning LLMs for LogQL. Vishwanath Seshagiri, Siddharth Balyan, **Vaastav Anand**, Kaustubh Dhole, Ishan Sharma, Avani Wildani, José Cambroneró, Andreas Züfle. In **arXiv 2025**, <https://arxiv.org/pdf/2412.03612>.
4. Columbo: Low Level End-to-End System Traces through Modular Full-System Simulation. Jakob Görgen, **Vaastav Anand**, Hejing Li, Jialin Li, and Antoine Kaufmann. In **arXiv 2024**, <https://arxiv.org/pdf/2408.05251>.
5. SoK: The Faults in our Graph Benchmarks. Puneet Mehrotra*, **Vaastav Anand***, Daniel Margo, Milad Rezaei Hajidehi, and Margo Seltzer. In **arXiv 2024**, <https://arxiv.org/pdf/2404.00766>.
6. Aggregate-driven trace visualizations for performance debugging. **Vaastav Anand**, Matheus Stolet, Thomas Davidson, Ivan Beschastnikh, Tamara Munzner, and Jonathan Mace. In **arXiv 2020**, <https://arxiv.org/pdf/2010.13681>.

Datasets

1. X-Trace distributed trace dataset for DeathStarBench. **Vaastav Anand** and Jonathan Mace. https://gitlab.mpi-sws.org/cld/trace-datasets/deathstarbench_traces
2. Fantasy Premier League Gameweek-By-Gameweek Dataset. Vaastav Anand. <https://github.com/vaastav/Fantasy-Premier-League>

Teaching

Instructor, Saarland University	2025
• Reliability in Modern Cloud Systems	2025
Teaching Assistant, Saarland University	2021
• 1× TA for Distributed Systems (Core Course)	2021
Graduate Teaching Assistant	2018-2020
University of British Columbia, Department of Computer Science	
• 1× TA for Computer Hardware and Operating Systems (CPSC 313)	2020
• 1× TA for Graduate Operating Systems (CPSC 508)	2019
• 1× TA for Distributed Systems (CPSC 416)	2018
Academic Assistant	2018
Vancouver Summer Program	
• Teaching Assistant for the Algorithms and the World Wide Web course.	
Undergraduate Teaching Assistant	2014-2018
University of British Columbia, Department of Computer Science	
• 1× TA for Introduction to Software Engineering (CPSC 210)	2018
• 1× TA for Advanced Operating Systems (CPSC 415)	2017

	• 1× TA for Intermediate Algorithm Design and Analysis (CPSC 320) 2017 • 1× TA for Computer Hardware and Operating Systems (CPSC 313) 2016 • 1× TA for Introduction to Computer Systems (CPSC 213) 2015 • 3× TA for Models of Computation (CPSC 121) 2014-2015
Awards	Busy Beaver Award, Best Seminar in CS, Saarland University 2026 3rd Place, Graduate Category, ACM SRC Grand Finals 2025 1st Place, Graduate Category, SOSP'24 SRC 2024 SoCC Student Scholarship 2019 2nd Place, Undergraduate Category, FSE'18 SRC 2018 SIGSOFT CAPS Award 2018 UBC International Tuition Award 2018-2019 Work Learn International Undergraduate Research Award 2018 UBC Faculty of Science, International Student Award 2015, 2018 ACM ICPC PacNW Regional Contest Division 2 Champion 2017 UBC Trek Excellence Scholarship 2016-17, 2017-2018 UBC Dean's Honor List 2014, 2015, 2017 UBC Computer Science Student Service Award 2015 GHS Global Citizen Scholarship 2011-2013
Service	Academic Service • Program Committee Member – SoCC 2026 – EuroSys Posters 2025 – CS-Can Student Symposium 2019 • Organizer – Tutorial on using Blueprint to accelerate Microservice Research @ SOSP'24 – Panel on Reproducibility and Replication @ HotOS'23 • Publicity Chair – The Journal of Systems Research 2022-2023. • Systems Trivia Co-Organizer – HotOS 2023 – SOSP 2021 – HotOS 2021 • Organization Committee Member – The Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS) 2022
Invited Talks	Blueprint: A Toolchain for Highly-Reconfigurable Microservices Jun 2024 Event: CNI Seminar Series @ IISc Bangalore Host: Sreeshma Shiv Blueprint: A Toolchain for Highly-Reconfigurable Microservices Jun 2024 Event: Azure Monitor Day of Learning @ Microsoft Redmond Host: Kalyana Sundaram

Blueprint: A Toolchain for Highly-Reconfigurable Microservices Aug 2023
Event: Trustworthy Systems Seminar @ UNSW Sydney
Host: Peter Chubb

Blueprint: A Toolchain for Highly-Reconfigurable Microservices Aug 2023
Event: Basser Seminar Series @ University of Sydney
Host: Zhanna Sarsenbayeva

Millenial: Modular Microservice Macrobenchmarks Jun 2021
Event: Tracing Jamboree @ Emory University (virtual)
Organizers: Avani Wildani, Ymir Vigfusson

Interests

Computing: Distributed Systems, Operating Systems, Software Engineering
Extra Curricular: Soccer, Linguistics, Cricket.